

5a	The field strength at a point equals the <u>negative</u> of the potential gradient there. i.e. the electric potential gradient is the electric field strength the <u>direction</u> of the field is the same as the direction of <u>decreasing</u> potential.	A	(B2) B1 B1
5bi	Straight line vertically upward	E	B1
5bii	$E = V/d$ $= 75/(1.2 \times 10^{-2})$ $= 6250 \text{ V m}^{-1}$	E	A1
5biii	gain in kinetic energy (= loss in potential energy) = charge $\times$ p.d. or $qV = \frac{1}{2}mv^2$ because separation not in expressions so v is independent of separation	D	B1 A0
5biv	(at $x = 0.40 \text{ cm}$), potential = $(-)\ 75 \times 0.40 / 1.2$ (= $(-)\ 25 \text{ V}$) $\frac{1}{2}mv^2 = qV$ $\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 25$ Or $a = Vq / dm$ and $v^2 = 2as$	D	C1 C1

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	$v^2 = (2 \times 75 \times 2 \times 1.60 \times 10^{-19} \times 0.40 \times 10^{-2}) / (1.2 \times 10^{-2} \times 4 \times 1.66 \times 10^{-27})$ $v = 4.9 \times 10^4 \text{ m s}^{-1}$		(C1)
			A1

6a	A progressive wave is a wave in which energy is carried from one point to another by	E	B1
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